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Inventor(s)	Nakamura; Hiroyuki et al.

High-performance and flexible loudspeaker apparatus usable in various scenes

Abstract

A high-performance and flexible loudspeaker apparatus usable in various scenes. A first unit includes a loudspeaker therein, a second unit is connected to a lower portion of the first unit, the first unit comprises the loudspeaker provided in a housing of the first unit so as to output a sound toward an upper portion of the housing and a passive radiator provided on a rear surface side of the loudspeaker in the housing, and an output port of the passive radiator is opposed to an inclined surface which is provided at an upper portion of the second unit and whose distance from the output port of the passive radiator increases toward a radially outer side of the housing.

Inventors: Nakamura; Hiroyuki (Tokyo, JP), Tsuruta; Minoru (Chichibu, JP), Yamada; Yoshihito (Honjo, JP), Kakizakai; Takao (Chichibu, JP)

Applicant: CANON DENSHI KABUSHIKI KAISHA (Saitama, JP)

Family ID: 86896502

Assignee: CANON DENSHI KABUSHIKI KAISHA (Saitama, JP)

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Primary Examiner: Chin; Vivian C

Assistant Examiner: Kang; Annabelle

Attorney, Agent or Firm: VENABLE LLP

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a loudspeaker apparatus such as a portable wireless loudspeaker.

Description of the Related Art

(2) In recent years, there has been proposed a wireless loudspeaker in which a lighting apparatus is built in an omnidirectional wireless loudspeaker emitting a sound in 360-degree directions to

express emotionality of music by means of dim light or to change the light volume along with music (refer to, for example, Japanese Laid-Open Patent Publication (kokai) No. 2016-19129).

(3) Each of the aforementioned conventional wireless loudspeakers is not intended to be used as a spotlight for illuminating the food or the like on the table, or used when the user reads or works at the desk. However, it is also conceivable that the loudspeaker is usable by increasing the light volume not only as a wireless loudspeaker but also as a lighting device at the time of reading or the like.

(4) Meanwhile, since the conventional wireless loudspeaker includes a low tone loudspeaker and a passive radiator on the bottom surface of the apparatus, there is a problem in which sound vibration may be transmitted to the stand such as a desk to generate a chattering sound, which may affect the sound quality.

SUMMARY OF THE INVENTION

(5) The present invention provides a high-performance and flexible loudspeaker apparatus usable in various scenes.

(6) Accordingly, the present invention provides a loudspeaker apparatus comprising a first unit including a loudspeaker therein, and a second unit connected to a lower portion of the first unit, the first unit comprising the loudspeaker provided in a housing of the first unit so as to output a sound toward an upper portion of the housing and a passive radiator provided on a rear surface side of the loudspeaker in the housing, an output port of the passive radiator being opposed to an inclined surface which is provided at an upper portion of the second unit and whose distance from the output port of the passive radiator increases toward a radially outer side of the housing.

(7) According to the present invention, it is possible to provide a loudspeaker apparatus with excellent sound quality that suppresses deterioration in sound quality caused by transmission of sound vibration to the stand such as a desk.

(8) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A to 1D are perspective views showing an example of a lighting-equipped loudspeaker according to a first embodiment of the present invention.

(2) FIGS. 2A to 2F are six-sided views showing the example of the lighting-equipped loudspeaker according to the first embodiment.

(3) FIG. 3 is a perspective cross-sectional view of the lighting-equipped loudspeaker according to the first embodiment.

(4) FIG. 4 is a cross-sectional view of the lighting-equipped loudspeaker according to the first embodiment.

(5) FIGS. 5A and 5B are bottom views showing examples of the attachment/detachment phases of a battery cover of the lighting-equipped loudspeaker according to the first embodiment.

(6) FIGS. 6A and 6B are perspective views of the lighting-equipped loudspeaker according to a second embodiment of the present invention as viewed diagonally from below.

(7) FIGS. 7A and 7B are perspective views for describing an internal configuration of a loudspeaker unit of the lighting-equipped loudspeaker according to the second embodiment.

(8) FIGS. 8A to 8E are diagrams for describing a configuration of a loudspeaker frame according to the second embodiment.

(9) FIGS. 9A and 9B are diagrams for describing a configuration of a loudspeaker substrate according to the second embodiment.

(10) FIGS. 10A and 10B are a front view and a cross-sectional view of the lighting-equipped

loudspeaker according to the second embodiment.

(11) FIGS. **11A** and **11B** are perspective views for describing a variation of an internal configuration of the loudspeaker unit of the lighting-equipped loudspeaker according to a third embodiment of the present invention.

(12) FIGS. **12A** to **12E** are diagrams for describing a variation of a configuration of a loudspeaker frame according to the third embodiment.

(13) FIGS. **13A** to **13E** are diagrams showing a use state by changing the position and the direction of a lighting unit according to a fourth embodiment of the present invention.

(14) FIGS. **14A** to **14E** are diagrams showing a use state by changing the position and the direction of the lighting unit according to the fourth embodiment.

(15) FIGS. **15A** to **15E** are diagrams showing a use state by changing the position and the direction of the lighting unit according to the fourth embodiment.

(16) FIGS. **16A** to **16F** are a front view and cross-sectional views of the lighting-equipped loudspeaker according to the fourth embodiment.

(17) FIGS. **17A** to **17D** are diagrams for describing an example of a configuration of a light shielding sheet according to the fourth embodiment.

DESCRIPTION OF THE EMBODIMENTS

(18) The present invention will now be described in detail below with reference to the accompanying drawings showing embodiments thereof.

First Embodiment

(19) First, a lighting-equipped loudspeaker **1** according to the present embodiment will be described with reference to FIGS. **1A** to **5B**. It should be noted that, as the lighting-equipped loudspeaker according to the present embodiment, a loudspeaker with a size of being usable on a table, a desk, or the like will be assumed, but the present invention is not limited thereto. For example, the lighting-equipped loudspeaker may be one usable by being connected to an instrument or the like in a live music club, an event venue, or the like.

(20) FIGS. **1A** to **1D** are perspective views showing an example of a lighting-equipped loudspeaker showing an embodiment of the present invention. FIG. **1A** is a perspective view of the lighting-equipped loudspeaker as viewed diagonally from the upper left, FIG. **1B** is a perspective view of the lighting-equipped loudspeaker as viewed diagonally from the upper right, FIG. **1C** is a perspective view of the lighting-equipped loudspeaker as viewed diagonally from the lower left, and FIG. **1D** is a perspective view of the lighting-equipped loudspeaker as viewed diagonally from the lower right.

(21) FIGS. **2A** to **2F** are six-side views showing the example of the lighting-equipped loudspeaker according to the present embodiment. In FIG. **2**, FIG. **2A** is a front view of the lighting-equipped loudspeaker, FIG. **2B** is a left side view of the lighting-equipped loudspeaker, FIG. **2C** is a right side view of the lighting-equipped loudspeaker, FIG. **2D** is a rear view of the lighting-equipped loudspeaker, FIG. **2E** is a plan view of the lighting-equipped loudspeaker, and FIG. **2F** is a bottom view of the lighting-equipped loudspeaker.

(22) FIG. **3** is a perspective cross-sectional view of the lighting-equipped loudspeaker according to the present embodiment. FIG. **4** is a cross-sectional view of the lighting-equipped loudspeaker according to the present embodiment. FIGS. **5A** and **5B** are bottom views showing examples of the attachment/detachment phases of a battery cover of the lighting-equipped loudspeaker according to the present embodiment.

(23) The lighting-equipped loudspeaker **1** according to the present embodiment includes a lighting unit **11**, a loudspeaker unit **12**, a battery unit **13**, and a rotation seat **14** from the top, and includes arms **15** on the right and left. The lighting unit **11** includes a light emitting unit (an LED light is used in the present embodiment, but the present invention is not limited thereto) therein, and emits light downward in the state of FIGS. **1A** to **1D** and FIGS. **2A** to **2F**.

(24) The loudspeaker unit **12** is an omnidirectional loudspeaker emitting a sound in 360-degree

directions. The loudspeaker unit **12** includes on the upper portion a mesh-like cover **121** and a loudspeaker **125** arranged upward, and on the lower portion a beam member **122**, a passive radiator **126** arranged downward, and the like. The beam member **122** is provided with a slit **123**. The loudspeaker unit **12** outputs a sound from the mesh-like cover **121** and the slit **123**. The loudspeaker unit **12** is connected to the battery unit **13** by the beam member **122**.

(25) The battery unit **13** has a below-mentioned battery, and supplies power to the lighting unit **11** and the loudspeaker unit **12**. The battery unit **13** is rotatable (or turnable) with respect to the rotation seat **14** with the rotation direction thereof in a horizontal plane (in a mounting plane of the apparatus).

(26) The arm **15** is rotatably supported by the battery unit **13**, and rotatably supports the lighting unit **11** at the tip end. The battery unit **13** includes a body **131** having substantially the same diameter as a body **124** of the loudspeaker unit **12**.

(27) As shown in FIGS. **3** and **4**, the battery unit **13** includes a rechargeable battery **132** and a substrate **133**. The substrate **133** is provided with a power supply circuit and the like. It should be noted that, as shown in FIG. **2D**, a USB terminal **138** is provided on the rear surface of the battery unit **13**, and power can be supplied to the battery **132** via the USB terminal **138**. It should be noted that the USB terminal **138** is, for example, type C, but may be another type of terminal. Also, data communication with an external device via the USB terminal **138** may be possible.

(28) A bottom portion **135** of the battery unit **13** has an opening for replaceably mounting the battery **132**, and a rectangular battery cover **136** is detachably attached to the opening. The battery **132** can be replaced by detaching the battery cover **136**. The battery unit **13** is rotatably supported by the rotation seat **14**. Hereinafter, description will be given with reference to FIGS. **5A** and **5B**.

(29) As shown in FIGS. **5A** and **5B**, the bottom portion **135** and the battery cover **136** rotate (or turn) in the horizontal direction with respect to the rotation seat **14**. The battery cover **136** is attached to the bottom portion **135** so as to rotate (or turn) about the intersection of the diagonals of the battery cover **136** at the time of the rotation.

(30) It should be noted that, in the present embodiment, in order to replaceably mount a larger battery, the size of the battery cover **136** (the length of a diagonal **136a** serving as a longest portion (corresponding to the rotation diameter)) is set to be larger than a diameter (**140a**) of the hole provided at the bottom portion of the rotation seat **14**. Then, only at the position shown in FIG. **5A**, the diameter of the hole is made long enough for the battery cover **136** to be detachable (to be in a shape of a portion **140b**).

(31) That is, when the rotation phase of the battery unit **13** with respect to the rotation seat **14** is the phase in FIG. **5A** or the phase rotated by 180 degrees from the phase in FIG. **5A**, the corners of the battery cover **136** can pass through the hole at the bottom portion of the rotation seat **14**. On the other hand, when the rotation phase of the battery unit **13** with respect to the rotation seat **14** is other than the phase of FIG. **5A** or the phase rotated by 180 degrees from the phase of FIG. **5A**, as in the example shown in FIG. **5B**, the corners of the battery cover **136** are caught in the hole of the bottom portion of the rotation seat **14** and cannot pass through the hole. Therefore, the battery cover **136** can be detached only in the phase of FIG. **5A** or the phase rotated by 180 degrees from FIG. **5A**. That is, the battery **132** can be replaced only in the phase of FIG. **5A** or the phase rotated by 180 degrees from FIG. **5A**.

(32) With such a configuration, it is possible to further increase the size of the battery cover **136**, and thus, it is possible to replaceably mount a larger battery. That is, in the lighting-equipped loudspeaker having the above-described configuration, by setting the size of the battery cover **136** (the length of the diagonal **136a** serving as a longest portion) to be larger than the diameter of the hole provided at the bottom portion of the rotation seat **14**, a larger battery can replaceably be mounted. As a result, the battery can be driven for a long time, and usability can be improved. Also, the area of the bottom surface portion of the rotation seat **14** can be made larger, and the stability when the lighting-equipped loudspeaker **1** is placed as a portable apparatus can be improved. Also,

since the battery cover **136** is prevented from coming off unless it is in a specific rotation phase, it is possible to reduce the chances that the battery cover **136** comes off due to an unexpected impact or the like applied during carrying.

(33) In the above configuration, the battery cover **136** is formed in a rectangular shape and is mounted at the position at which the intersection (**136b**) of the diagonals (**136a**) corresponds to the rotation center with respect to the rotation seat **14**, and the hole provided in the rotation seat **14** is in a shape having the shorter diameter portion (**140a**) shorter than the length of the diagonal of the battery cover **136** and the longer diameter portion (**140b**) longer than the length of the diagonal of the battery cover **136**. When the rotation seat **14** is in the specific rotation phase, the diagonal (**136a**) of the battery cover **136** is located at the longer diameter portion (**140b**) of the hole of the rotation seat **14**, and when the rotation seat **14** is not in the specific rotation phase, the diagonal (**136a**) of the battery cover **136** is located at the shorter diameter portion (**140a**) of the hole of the rotation seat **14**, so that the battery cover **136** can be detached (that is, the battery **132** can be replaced) only when the rotation seat **14** is in the specific rotation phase (in the case of FIG. 5A).

(34) However, the shape of the battery cover **136** is not limited to the rectangular shape, and may be another shape corresponding to the shape of the battery **132**. For example, the shape may be another polygonal shape, a circular shape that is not a perfect circle (a circular shape having a longer diameter portion and a shorter diameter portion), or the like. In this case, the hole provided in the rotation seat **14** is in a shape having a shorter diameter portion shorter than the rotation diameter of the battery cover **136** and a longer diameter portion longer than the rotation diameter of the battery cover **136**. When the rotation seat **14** is in a specific rotation phase, a portion corresponding to the rotation diameter of the battery cover **136** (that is, a portion on the outermost side with respect to the rotation center) is located in the longer diameter portion of the hole of the rotation seat **14**. When the rotation seat **14** is not in the specific rotation phase, the portion corresponding to the rotation diameter of the battery cover **136** is located in the shorter diameter portion of the hole of the rotation seat **14**. Accordingly, only when the rotation seat **14** is in the specific rotation phase, the battery cover **136** is made detachable, and the thus battery **132** is made replaceable.

(35) As described above, the lighting-equipped loudspeaker **1** according to the present embodiment is a vertical wireless loudspeaker, has a lighting unit capable of changing the position and the irradiation angle at the upper portion thereof, can be used not only as an audio device but also as a lighting device for reading or working, serves as an audio device and a lighting device, can be used for a longer time even in a place where it is difficult to secure a power source by mounting a large battery, and is suitable as, for example, a tabletop loudspeaker. It should be noted that, although the lighting-equipped loudspeaker has been described as an example, the present invention is also applicable to a loudspeaker apparatus (a loudspeaker apparatus including the loudspeaker unit **12**, the battery unit **13**, and the rotation seat **14**) not including the lighting unit **11**. In general, in a case where the light volume of a wireless lighting-equipped loudspeaker is set to light volume sufficient for reading, working, or the like, power consumption increases, and it is necessary to mount a large battery. Also, even in a case where it is used as a wireless loudspeaker, it is necessary to mount a large battery in order to enable long-time use. Also, the battery to be mounted is desirably replaceable without disassembling the apparatus. Since the conventional wireless loudspeaker is not intended for reading, working, or the like, it is not configured to be able to replaceably mount a large battery that can be used for such a purpose for a long time. It should be noted that it may be possible to replaceably mount a large battery by increasing the size of the apparatus, but such an apparatus has a problem in which the apparatus is unsuitable as, for example, a tabletop loudspeaker. On the contrary, the present embodiment has solved the problem as described above. Therefore, it is possible to provide a loudspeaker apparatus equipped with a large battery while preventing an increase in the external size as a wireless loudspeaker. Also, it is possible to provide a lighting-equipped loudspeaker that can be used not only as a wireless loudspeaker but also as a

lighting device at the time of reading or the like, and that can be used for a long time regardless of a place of use.

Second Embodiment

(36) As described above, the lighting-equipped loudspeaker **1** according to the present embodiment includes the loudspeaker unit **12** (first unit) having the loudspeaker **125** therein and the battery unit **13** (second unit) connected to the lower portion of the loudspeaker unit **12**. Then, as shown in FIGS. **3** and **4** described above, the lighting-equipped loudspeaker **1** includes the passive radiator **126** in the housing of the loudspeaker unit **12**. The passive radiator **126** operates using the aerial vibration generated in the housing when the loudspeaker **125** outputs a sound, and mainly amplifies and reinforces the low tone range. It should be noted that, in the loudspeaker unit **12**, the loudspeaker **125** is provided in the housing so as to output a sound toward the upper portion of the housing, and the passive radiator is further provided on the rear side (lower portion) of the loudspeaker **125** in the housing.

(37) FIGS. **6A** and **6B** are perspective views of the lighting-equipped loudspeaker **1** as viewed diagonally from below. FIG. **6A** is a diagram of the lighting-equipped loudspeaker **1** as viewed from the lower front, and FIG. **6B** is a diagram of the lighting-equipped loudspeaker **1** as viewed diagonally from the lower right. A sound output from the passive radiator **126** is output from the slit **123**.

(38) FIGS. **7A** and **7B** are perspective views for describing the internal configuration of the loudspeaker unit **12**. FIG. **7A** is a diagram of the internal configuration of the loudspeaker unit **12** as viewed diagonally from above, and FIG. **7B** is a diagram of the internal configuration of the loudspeaker unit **12** as viewed diagonally from below. The loudspeaker **125** is supported by a loudspeaker frame **12f**. A sound absorbing material **128** is provided inside the loudspeaker frame **12f**. The sound absorbing material **128** is a member for absorbing a sound inside the loudspeaker unit **12**. The sound absorbing material **128** has, for example, the effect of flattening frequency characteristics by absorbing medium or high pitched sounds and standing waves, and the effect of lowering resonance sharpness. The lower portion of the sound absorbing material **128** is supported by a resin member **129** (FIGS. **8A** to **8E**). The sound absorbing material **128** is made of an elastic body and is elastically deformed. The sound absorbing material **128** is attached between the loudspeaker **125** and the resin member **129** in a deformed state. The attached sound absorbing material **128** tries to return to its original shape due to elastic deformation, and a frictional force is generated between the sound absorbing material and contact portions with the loudspeaker **125** and the resin member **129**. Due to this frictional force, the sound absorbing material **128** is held between the loudspeaker **125** and the resin member **129**.

(39) FIGS. **8A** to **8E** are diagrams for describing a configuration of the loudspeaker frame **12f**. FIG. **8A** is a diagram of the loudspeaker frame **12f** as viewed from directly below. FIGS. **8B** and **8C** are perspective views of the loudspeaker frame **12f** as viewed diagonally from below. FIG. **8D** is a side view of the loudspeaker frame as viewed in the direction indicated by the arrow B in FIG. **8A**. FIG. **8E** is a side view of the loudspeaker frame as viewed in the direction indicated by the arrow C in FIG. **8A**.

(40) The resin member **129** shown in FIGS. **8A** to **8E** is a support member made of resin and supports the sound absorbing material **128** from below. The resin member **129** is in a shape having a step **129g**, and has a configuration capable of sandwiching the sound absorbing material **128** between the resin member and the loudspeaker **125** while avoiding interference with a capacitor **1271** (FIGS. **9A** and **9B**) due to the step **129g**. Also, the resin member **129** has one side thereof formed in a largely cut-out shape, and this cut-out portion forms a sound passage **120** in the housing of the loudspeaker unit **12**. Through this sound passage **120**, aerial vibration (a sound) from the upper portion of the loudspeaker unit **12** is transmitted to the passive radiator **126** provided on the lower side.

(41) FIGS. **9A** and **9B** are diagrams for describing a configuration of a loudspeaker substrate **127**.

FIG. 9A is a side view of the loudspeaker substrate **127**, and FIG. 9B is a plan view of the loudspeaker substrate **127** as viewed from above. The loudspeaker substrate **127** is provided with various electronic components which are not shown, the capacitor **1271**, a wireless module (for example, a Bluetooth (registered trademark) module **1272**), and the like. It should be noted that the wireless module may be another wireless module such as a Wi-Fi module. The loudspeaker substrate **127** is formed in a cut-out circular shape such as a semicircle (a circle in which a part (left side in FIG. 9B) is cut out), and this cut-out portion serves as a sound passage. Through this sound passage, aerial vibration (the sound) from the upper portion of the loudspeaker unit **12** is transmitted to the passive radiator **126** provided on the lower portion.

(42) FIGS. 10A and 10B are a front view and a cross-sectional view of the lighting-equipped loudspeaker **1**. FIG. 10A is a front view of the lighting-equipped loudspeaker **1**, and FIG. 10B is a cross-sectional view of the lighting-equipped loudspeaker **1** cut along the cut surface A-A shown in FIG. 10A. As shown in FIG. 10B, a space is formed between the inner surface of the housing of the loudspeaker unit **12** and the periphery of the sound absorbing material **128**. That is, there is a gap around the sound absorbing material **128**, and spaces existing between the loudspeaker **125** and the passive radiator **126** linearly communicate with each other through the gap. This space serves as a sound passage in the loudspeaker unit **12**.

(43) As shown in each of the above-described figures, in the lighting-equipped loudspeaker **1**, the passive radiator **126** is provided not at the bottom portion of the lighting-equipped loudspeaker **1** in contact with a stand such as a desk, but in the loudspeaker unit **12** in the middle not in contact with the stand. As a result, the vibration of a sound output from the passive radiator **126** is less likely to be transmitted to the stand. Further, an upper surface **13t** of the battery unit **13** is located at a position opposed to the output port (slit **123**) of the passive radiator **126**. The upper surface **13t** is an inclined surface whose distance from the output port of the passive radiator **126** increases toward the radially outer side of the housing. The upper surface **13t** and the side surface of the battery unit **13** are made of metal such as strong aluminum having an integrated structure, and can appropriately reflect a sound while suppressing generation of a chattering sound due to the sound output from the passive radiator **126**. In this manner, in the lighting-equipped loudspeaker **1**, it is possible to suppress deterioration in sound quality caused by a chattering sound or the like. It should be noted that the metal constituting the housing of the battery unit **13** is not limited to aluminum, and may be other kinds of metal.

(44) Since the conventional wireless loudspeaker includes the passive radiator on the bottom surface of the apparatus, there is a problem in which sound vibration may be transmitted to the stand to generate a chattering sound, which may deteriorate the sound quality. However, the lighting-equipped loudspeaker **1** solves this problem.

(45) Also, as described above, the resin member **129** is in a shape having the step **129g**, and is attached so that the capacitor **1271** on the loudspeaker substrate **127** may be housed in the space created by the step **129g**. That is, the resin member **129** supports the sound absorbing material **128** while avoiding interference with the capacitor **1271**. With such a configuration, downsizing of the loudspeaker unit **12** is achieved (an increase in size can be suppressed). Furthermore, the loudspeaker unit **12** includes, in the housing thereof, the loudspeaker **125** provided so as to output a sound toward the outside of the housing (upward), the sound absorbing material **128** on the rear surface side (lower side) of the loudspeaker **125**, the loudspeaker substrate **127** formed in a cut-out circular shape on the side (lower side) of the sound absorbing material **128** opposite to the loudspeaker **125**, and the passive radiator **126** provided on the side of the loudspeaker substrate **127** opposite to the sound absorbing material **128** so as to output a sound toward the outside of the housing (downward).

(46) It should be noted that the sound absorbing material **128** is provided so as to form a space (gap) around the sound absorbing material **128**, that is, between the sound absorbing material **128** and the inner surface of the housing of the loudspeaker unit **12**, and this gap serves as a sound

passage. Specifically, as shown in FIGS. 7A and 7B, the sound absorbing material **128** is formed in a rectangular solid shape, and a gap is formed between each side and the inner surface of the circular housing.

(47) Also, the Bluetooth (registered trademark) module **1272** is provided in the cut-out portion of the cut-out circular loudspeaker substrate **127** (FIGS. 9A and 9B). It should be noted that the cut-out portion of the cut-out circular loudspeaker substrate **127** also serves as a sound passage. With such a configuration of forming a sound passage, aerial vibration (a sound) from the loudspeaker **125** provided at the upper portion of the loudspeaker unit **12** is appropriately transmitted to the passive radiator **126** without increasing the size. In particular, in the present embodiment, there is a positional relationship in which the gap between each side of the sound absorbing material **128** and the inner surface of the housing and the cut-out portion of the cut-out circular loudspeaker substrate **127** overlap with each other in the vertical direction, and a sound is more easily transmitted from the loudspeaker **125** to the passive radiator **126**.

(48) As described above, according to the second embodiment, it is possible to provide a compact loudspeaker apparatus that achieves excellent sound quality while suppressing an increase in size of the apparatus.

Third Embodiment

(49) FIGS. 11A and 11B are perspective views for describing a variation of the internal configuration of the loudspeaker unit **12**. FIG. 11A is a diagram of the variation of the internal configuration of the loudspeaker unit **12** as viewed diagonally from above, and FIG. 11B is a diagram of the variation of the internal configuration of the loudspeaker unit **12** as viewed diagonally from below. A loudspeaker frame **12g** is disposed between the loudspeaker **125** and the passive radiator **126**. The loudspeaker **125** is supported by the loudspeaker frame **12g**, and the passive radiator **126** is suspended and supported by the loudspeaker frame **12g**. A sound absorbing material **200** is attached to the loudspeaker frame **12g**. The sound absorbing material **200** is a member for absorbing a sound inside the loudspeaker unit **12**. Similarly to the sound absorbing material **128**, the sound absorbing material **200** is made of an elastic body, and exhibits, for example, the effect of flattening frequency characteristics by absorbing medium or high pitched sounds and standing waves, and the effect of lowering resonance sharpness. The sound absorbing material **200** is attached to the loudspeaker frame **12g** (FIGS. 12A to 12E) so as to cover a part of the semicircular portion in the cross section of the substantially cylindrical space between the loudspeaker **125** and the passive radiator **126**.

(50) FIGS. 12A to 12E are diagrams for describing a configuration of the loudspeaker frame **12g**. FIG. 12A is a diagram of the loudspeaker frame **12g** as viewed from directly above. FIGS. 12B and 12C are perspective views of the loudspeaker frame **12g** as viewed diagonally from above. FIG. 12D is a side view of the loudspeaker frame as viewed in the direction indicated by the arrow D in FIG. 12A. FIG. 12E is a side view of the loudspeaker frame as viewed in the direction indicated by the arrow E in FIG. 12A.

(51) The loudspeaker frame **12g** is a frame body protruding downward and formed substantially in a U shape in a side view, and a substantially semicircular substrate **201** is attached thereto. The substrate **201** is disposed so that the surface thereof is parallel to the vertical direction (arrow direction in FIG. 11A) from the loudspeaker **125** toward the passive radiator **126**. At this time, the circumferential portion of the substrate **201** protrudes downward and substantially coincides with the bent portion of the bottom portion of the loudspeaker frame **12g** (FIG. 11A). Furthermore, the sound absorbing material **200** is disposed so as to sandwich the loudspeaker frame **12g** between the sound absorbing material **200** and the substrate **201** (FIG. 11B).

(52) Since the substrate **201** is disposed so that the surface thereof is parallel to the vertical direction, the space between the loudspeaker **125** and the passive radiator **126** directly communicates to form a sound passage **203** in the housing of the loudspeaker unit **12**. As described above, the sound absorbing material **200** closes only a part of the semicircular portion in the cross

section of the space between the loudspeaker **125** and the passive radiator **126**. Therefore, the cross-sectional area of the sound passage **203** is larger than the cross-sectional area of the sound passage according to the second embodiment formed by the gap between the inner surface of the housing of the loudspeaker unit **12** and the periphery of the sound absorbing material **128**. As a result, since the aerial vibration (a sound) from the upper portion of the loudspeaker unit **12** is efficiently transmitted to the passive radiator **126**, a sound in the low tone range of the loudspeaker **125** can further be amplified and reinforced by the passive radiator **126** than in the second embodiment.

(53) Also, similarly to the loudspeaker substrate **127**, the substrate **201** is provided with various electronic components which are not shown, a wireless module (for example, a Bluetooth (registered trademark) module **202**), and the like. It should be noted that the wireless module may be another wireless module such as a Wi-Fi module. In particular, the Bluetooth module **202** is disposed on the substrate **201** so as to face the sound absorbing material **200**, but a part thereof protrudes upward from the substrate **201** and is exposed to the sound passage **203**. As described above, since the cross-sectional area of the sound passage **203** is large, it is easy to perform wireless communication between the Bluetooth module **202** and a wireless communication device outside the lighting-equipped loudspeaker **1** via the sound passage **203**. This enables various operations of the lighting-equipped loudspeaker **1** to be reliably performed by an external wireless communication device via the Bluetooth module **202**.

Fourth Embodiment

(54) FIGS. **13A** to **15E** are diagrams showing states in which the loudspeaker apparatus (lighting-equipped loudspeaker **1**) according to the present embodiment is used by changing the position and the direction of the lighting unit **11**. The lighting-equipped loudspeaker **1** according to the present embodiment includes the loudspeaker unit **12** (first unit) having the loudspeaker **125** therein, the lighting unit **11** provided on the upper side of the loudspeaker unit **12**, and the battery unit **13** (second unit) connected to the lower portion of the loudspeaker unit **12**. The lighting unit **11** includes a light emitting unit (an LED light (a below-mentioned LED substrate **114**) is used in the present embodiment, but the present invention is not limited thereto) therein, and emits light downward in the state of FIGS. **1A** to **1D** and FIGS. **2A** to **2F**. In the lighting-equipped loudspeaker **1**, a hinge **137** (FIGS. **3** and **4**) is provided in the battery unit **13** and turnably supports the arm **15**. Also, in the lighting-equipped loudspeaker **1**, a hinge **113** (FIGS. **3** and **4**) is provided in the lighting unit **11** and turnably supports the lighting unit **11**. With such a configuration, in the lighting-equipped loudspeaker **1**, the arm **15** and the lighting unit **11** can freely turn. Therefore, the user can use the lighting-equipped loudspeaker **1** as a lighting device at the time of reading, working, or the like by illuminating an arbitrary place that the user wishes to be illuminated.

(55) FIGS. **16A** to **16F** are a front view and cross-sectional views of the lighting-equipped loudspeaker **1**. FIG. **16A** is a front view of the lighting-equipped loudspeaker **1**, and FIG. **16B** is a cross-sectional view of the lighting-equipped loudspeaker **1** cut along the cut surface A-A shown in FIG. **16A**. FIG. **16C** is a cross-sectional view of the lighting-equipped loudspeaker **1** cut along the cut surface B-B shown in FIG. **16A**, and FIG. **16D** is an enlarged view of the portion DT1 of the cut surface shown in FIG. **16C**, and corresponds to a cross-sectional view of a portion to which the arm **15** is connected. Also, FIG. **16E** is a cross-sectional view of the lighting-equipped loudspeaker **1** cut along the cut surface C-C shown in FIG. **16A**, and FIG. **16F** is an enlarged view of the portion DT2 of the cut surface shown in FIG. **16E**, and corresponds to a cross-sectional view further on the upper portion than the portion to which the arm **15** is connected.

(56) As shown in FIGS. **16A** and **16B**, a diffuser **115** (diffusion plate) is provided on the lower portion of the lighting unit **11**. The diffuser **115** is made of a light transmissive, milky white material, for example, and is formed in a downward (that is, the direction in which the lighting unit **11** emits light) substantially conical shape (tapered shape) as shown in FIGS. **16A** and **16B**, and this tapered shape has the effect of diffusing the irradiation light from the light emitting unit **114** in

all directions.

(57) Also, in a case where the diffuser **115** takes a posture in which the center of the diffuser **115** and the center of the loudspeaker **125** are opposed to each other and are substantially aligned as shown in FIGS. **16A** and **16B**, the diffuser **115** has the effect of diffusing the irradiation light from the light emitting unit **114** in all directions while preventing front reflection of a sound output from the loudspeaker **125** due to the above-described tapered shape. Also, as shown in FIG. **14C**, the diffuser **115** may be formed in a shape in which an angle of the tapered shape changes at a boundary **115a** such as to diffuse light and a sound at an optimum angle. In this case, the center side area inside the boundary **115a** may be a flat surface having no diffusion effect that does not affect the reflection of a sound. Also, a camera may be provided inside the lighting unit **11** such that a lens of the camera is arranged on the center side area inside the boundary **115a** or another place. Further, an audio input unit (not shown) having a microphone therein may be provided on the front surface of the lighting unit **11**.

(58) Also, as shown in FIGS. **13A** to **15E**, in the lighting-equipped loudspeaker **1** according to the present embodiment, for example, the arm **15** is turnable in the range of -25 degrees to 25 degrees (angle in FIG. **13B**) with the vertical direction as 0 degrees, the lighting unit **11** is turnable in the range of -90 degrees (angle in FIGS. **14A** to **14E**) to 180 degrees (angle in FIGS. **15A** to **15E**) with the state shown in FIG. **2** as 0 degrees, and the rotation seat **14** is turnable by 360 degrees. Also, the illumination by means of the lighting unit **11** can be controlled, and for example, sunlight colors and warm colors can be switched with a plurality of levels of brightness. Also, in the lighting-equipped loudspeaker **1** according to the present embodiment, the position and the irradiation angle of the lighting unit **11** provided at the upper portion of the apparatus can easily be changed with one hand without the need for support of the apparatus main body with the other hand, and light can freely be emitted in an arbitrary direction that the user wishes for. It should be noted that changing the position and the irradiation angle of the lighting unit **11** as described above causes the reflection, the directivity, and the quality of a sound output from the loudspeaker **125** to be changed. That is, the user can change the sound quality by changing the position and the irradiation angle of the lighting unit **11**. It should be noted that, since the lighting unit **11** is formed in a cylindrical shape, and the diffuser **115** is formed in a conical shape, the output port of the loudspeaker **125** and the surface of the lighting unit **11** do not face each other even when the angle of the lighting unit **11** is set to any angle from -90 degrees to 90 degrees.

(59) Also, the lighting unit **11** may be used in a manner in which the angle of the lighting unit **11** is adjusted in a range of -90 degrees shown in FIGS. **14A** to **14E** to 180 degrees shown in FIGS. **15A** to **15E**, and in which a sound output from the loudspeaker **125** is reflected in a specific direction on the upper surface of the lighting unit **11**, so that the loudspeaker **125** may have directivity. By providing directivity, for example, it is possible to improve listening easiness of a conversation in a video conference (TV conference) or the like. In this manner, the user can change the sound quality and the directivity in accordance with the use application such as music appreciation and video conferencing.

(60) As described above, in the lighting-equipped loudspeaker **1** according to the present embodiment, the arm **15** is turnably attached to the lighting unit **11** in order to achieve the configuration in which the position and the irradiation angle of the lighting unit **11** can freely be changed. The lighting unit **11** is provided with a hole through which the arm **15** passes at an attachment portion of the arm **15**, and the arm **15** is inserted into the hole. Therefore, there is a concern that light of the light emitting unit (LED substrate **114**) inside the lighting unit **11** may leak from the attachment portion of the arm **15** (the gap between the hole and the arm **15**). In order to prevent this, in the present embodiment, as shown in FIGS. **16D** and **16F**, a light shielding sheet **900** as shown in FIGS. **17A** to **17D** is provided at the attachment portion of the arm **15** in the lighting unit **11**.

(61) The light shielding sheet **900** is a ring-shaped light shielding sheet having a through hole

having a diameter substantially equal to the diameter of the arm **15**. The light shielding sheet **900** is disposed so as to be curved along the inner surface of the cylindrical housing of the lighting unit **11**, lets the arm **15** pass through the through hole thereof, and is sandwiched between the inner surface of the housing of the lighting unit **11** and the connection portion of the tip end of the arm **15**. At this time, the inner diameter of the through hole corresponds to the outer diameter of the arm **15** so that a gap may not be created between the inside of the ring of the light shielding sheet **900** and the arm **15**. It should be noted that FIGS. **16D** and **16F** show a state in which the light shielding sheet **900** is attached to the arm attachment portion of the lighting unit **11** on the right as viewed from the front, but similarly, another light shielding sheet **900** is also attached to an arm attachment portion on the left as viewed from the front.

(62) Hereinbelow, the light shielding sheet **900** will be described. FIGS. **17A** to **17D** are diagrams for describing an example of a configuration of the light shielding sheet **900**. FIG. **17A** is a perspective view of the light shielding sheet **900**, FIG. **17B** is a plan view of the light shielding sheet **900**, FIG. **17C** is a side view of the light shielding sheet **900**, and FIG. **17D** is a cross-sectional view of the light shielding sheet **900**.

(63) As shown in FIGS. **17A** and **17B**, the light shielding sheet **900** is formed in a ring shape, and is disposed so that the arm **15** will pass through the inside of the ring with no gap therebetween. As shown in FIG. **17C**, the light shielding sheet **900** includes a single-sided PET sheet attached cushion material **901** and a light shielding tape **902**. It should be noted that PET is an abbreviation for polyethylene terephthalate.

(64) Further, as shown in FIG. **17D**, the single-sided PET sheet attached cushion material **901** includes a PET film **901-1** and a polyurethane foam **901-2**. Also, the light shielding tape **902** includes a black polyester film **902-1** and a black acrylic adhesive **902-2**, and is attached to the single-sided PET sheet attached cushion material **901** by the adhesive force of the black acrylic adhesive **902-2**.

(65) It should be noted that, since a portion of the lighting unit **11** to which the arm **15** is attached is a movable portion, smooth movement is required. Also, the light shielding sheet **900** arranged at the attachment portion of the arm **15** is arranged in a state of being sandwiched between the inner surface of the housing of the lighting unit **11** and the connection portion of the tip end of the arm **15**. Therefore, by making the surface of the black polyester film **902-1** of the light shielding sheet **900** smooth and enhancing the slidability, smoother movement can be achieved. Further, the light shielding tapes **902** having high slidability may be attached to both the surfaces of the single-sided PET sheet attached cushion material **901** to further enhance the slidability and the light shielding property. That is, the light shielding sheet **900** may include the cushion material **901** and sheets (light shielding tapes **902**) having higher slidability than the cushion material **901** attached to both the surfaces of the cushion material **901**.

(66) In this manner, by attaching the ring-shaped light shielding sheet **900**, which can be curved along the inner surface of the housing of the lighting unit **11** and into which the arm **15** can be inserted with no gap therebetween, along the inner surface of the housing of the lighting unit **11**, to the attachment position of the arm **15** on the inner surface of the housing, it is possible to prevent light of the light emitting unit inside the lighting unit **11** from leaking from the attachment portion of the arm **15**.

(67) It should be noted that the lighting-equipped loudspeaker **1** according to the present embodiment has a configuration in which the arm **15** for changing the position and the irradiation angle of the lighting unit **11** is connected to the battery unit **13** and the lighting unit **11**, and the arm **15** is away from the loudspeaker unit **12** (not in contact with the loudspeaker unit **12**). That is, for example, the U-shaped arm **15** does not contact the loudspeaker unit **12** and has one end turnably supported by the battery unit **13** and the other end turnably support the lighting unit **11**. This can prevent the vibration of the loudspeaker unit **12** from being transmitted to the lighting unit **11** via the arm **15** and causing flickering or the like. In particular, in a case where the lighting unit **11**

includes a camera or a microphone, vibration transmitted to the lighting unit **11** may affect the image quality of an image shot by the camera or the sound quality of a sound input into the microphone. However, the present embodiment can suppress the influence of the vibration on the image quality and the sound quality.

(68) As described above, it is possible to provide an easy-to-use lighting-equipped loudspeaker that can be used not only as a wireless loudspeaker but also as a lighting device at the time of reading or the like by illuminating an arbitrary place that the user wishes to be illuminated, that is portable, and that has both the maintenance of the posture of the desk-lamp-type lighting device and the operability enabling easy movement with one hand without the need for support of the apparatus. In this manner, it is possible to provide a high-performance and flexible lighting-equipped loudspeaker apparatus usable in various scenes so as to be used to illuminate an arbitrary place that the user wishes to be illuminated or to reproduce a high-quality sound source or voice.

(69) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(70) This application claims the benefit of Japanese Patent Application No. 2021-211330, filed on Dec. 24, 2021, the benefit of Japanese Patent Application No. 2022-125555, filed on Aug. 5, 2022, and the benefit of Japanese Patent Application No. 2022-125556, filed on Aug. 5, 2022, and the benefit of Japanese Patent Application No. 2022-190372, filed on Nov. 29, 2022, which are hereby incorporated by reference herein in their entirety.

Claims

1. A loudspeaker apparatus comprising: a first unit including a loudspeaker therein; and a second unit connected to a bottom of the first unit, wherein the first unit comprises: the loudspeaker provided in a housing of the first unit so as to output a sound toward an upper portion of the housing; and a passive radiator provided on a bottom side of the loudspeaker in the housing, and wherein an output port of the passive radiator faces an inclined surface which is provided at an upper portion of the second unit and whose distance from the output port of the passive radiator increases toward a radially outer side of the housing.
2. The loudspeaker apparatus according to claim 1, wherein an upper surface of the second unit facing the passive radiator is made of metal.
3. The loudspeaker apparatus according to claim 1, wherein a sound absorbing material is provided between the loudspeaker and the passive radiator in the housing, a substrate formed in a cut-out circular shape is provided between the sound absorbing material and the passive radiator in the housing, and the sound absorbing material forms a gap between a periphery thereof and an inner surface of the housing, and spaces existing between the loudspeaker and the passive radiator linearly communicate with each other through the gap.
4. The loudspeaker apparatus according to claim 3, wherein a wireless module is attached to the cut-out portion of the substrate.
5. The loudspeaker apparatus according to claim 1, wherein a frame body is provided between the loudspeaker and the passive radiator in the housing, and a substrate is attached to the frame body so that a surface of the substrate is parallel to a direction from the loudspeaker toward the passive radiator to form a sound passage between the loudspeaker and the passive radiator.
6. The loudspeaker apparatus according to claim 5, wherein a sound absorbing material is attached to the frame body, and the sound absorbing material is disposed so as to sandwich the frame body between the sound absorbing material and the substrate.
7. The loudspeaker apparatus according to claim 5, wherein a wireless module is attached to the substrate, and a part of the wireless module is exposed to the sound passage.

8. A loudspeaker apparatus comprising: a first unit including a loudspeaker therein; and a second unit connected to a lower portion of the first unit, wherein the first unit comprises: the loudspeaker provided in a housing of the first unit so as to output a sound toward an upper portion of the housing; and a passive radiator provided on a rear surface side of the loudspeaker in the housing, and wherein an output port of the passive radiator is opposed to an inclined surface which is provided at an upper portion of the second unit and whose distance from the output port of the passive radiator increases toward a radially outer side of the housing, wherein a sound absorbing material is provided between the loudspeaker and the passive radiator in the housing, a substrate formed in a cut-out circular shape is provided between the sound absorbing material and the passive radiator in the housing, and the sound absorbing material forms a gap between a periphery thereof and an inner surface of the housing, and spaces existing between the loudspeaker and the passive radiator linearly communicate with each other through the gap, and wherein a support member made of resin that supports the sound absorbing material is provided between the sound absorbing material and the substrate in the housing, a predetermined electronic component is attached to the substrate on a side which the sound absorbing material faces, and the support member is in a shape having a step and is attached inside the housing so as for the predetermined electronic component to be housed in a space created by the step.
